

Examiner reports

Q1.

Students did well here, with over 80% of students identifying the correct option. There was no clear second choice.

Q2.

The majority recalled that integration was required in part (a), although some differentiated and a few used $s = vt$. A large proportion either forgot the constant of integration or did not calculate its value, losing half the marks.

In part (b), the time of 2 seconds ensured that the toy was stationary when it launched the ball. Many, however, tried to bring extraneous distances and velocities into this question. Making the model fit the situation is an important skill to be assessed. For those who did this, the actual calculation was usually carried out correctly.

Q3.

Nearly all students made good progress with part (a), with over half scoring full marks. Nearly all students successfully found the maximum possible friction and made a comparison with the applied horizontal force. The most frequently seen error was when students wrote a statement that was incorrect, often along the lines of “friction is greater than 150N so there is equilibrium”. This showed a misunderstanding of the friction model and could not be overlooked.

Part (b) was also well answered, but a lack of detail often meant that students had not fully justified their answer. Several impressive and detailed answers were seen, where students explained clearly that the horizontal component of the applied force was greater than the maximum friction so that the crate would not remain stationary.

Resolving of forces was generally successful.

Q9.

Part (a) was a very standard question and many students did well with this part of the question. However, a small number of students ignored the instruction to form two equations of motion. In part (b), some students did not convert the 40 cm to metres and obtained speeds that related to moving 40 m rather than 40 cm. These students did gain some marks, but lost the final accuracy mark.

In part (c) a common error was to use the wrong acceleration, with some students continuing to use 4.9 m s^{-2} rather than -9.8 m s^{-2} . In this part there was also some confusion between the values to use for u and v , with some students taking u as zero at the maximum height.

Most students were able to say that the acceleration would be reduced, when answering part (b), but a much smaller proportion of the students could give a good explanation. Those who mentioned the effect of the air resistance on the resultant force normally gained both marks.

Q10.

Part (a) was often done very well by students, with a reasonable number of students

gaining all 11 marks. There were a few issues with the force diagram, with missing or additional forces. Many students seemed to have been helped by the answer that was given for the normal reaction and this in turn enabled many to find the friction correctly. Finding the acceleration was also done well, although there were a few sign errors and some students simply used 150 instead of $150 \cos 20^\circ$. Part (b) was much more demanding and there were far fewer correct responses. The most common error was not to take account of the changed angle when finding the normal reaction and to simply continue to use the value of 243 from part (a).

In part (c) there were some good responses, but a fair number of students incorrectly stated that the tension would be greater for a greater constant speed.

Q11.

This question was generally done well. There were very many good force diagrams, with the most common error probably being the omission of arrow heads. Part (b) was done very well. While there were many good answers to part (b) some candidates simply calculated $4 \times 3 = 12$. There were also a few candidates who made a sign error, producing equations such as $F - 50 = 12$. Candidates were generally able to apply $F = \mu R$ to their answers from the previous parts, and many gained the follow through marks that were available. There were many good explanations for part (c), but also a number of confused answers, in which it was claimed that the coefficient of friction would have to increase.

Q12.

This question was done very well by candidates and there were many full correct solutions. The main source of errors in part (a) was due to using incorrect combinations of sin or cos and the angle that the candidates decided to work with. In part (b) some candidates used trigonometry while other used Pythagoras' Theorem. In the final part some candidates multiplied by 9.8 rather than dividing by it.

Q13.

The candidates who understood how to approach this question did very well, but others were unable to make much progress with this question.

Q14.

The candidates made very little progress with this question and did not seem to be able to deal with the resolving required.

Q15.

This question was done very well by candidates and there were many full correct solutions. The main source of errors in part (a) was due to using incorrect combinations of sin or cos and the angle that the candidates decided to work with. In part (b) some candidates used trigonometry while other used Pythagoras' Theorem. In the final part some candidates multiplied by 9.8 rather than dividing by it.

Q16.

Part (a) was done well by some candidates, but there were a significant number who proceeded as if this was a two particles connected by a string over a peg type of problem. Many times a pair of equations such as $18g - T = 18a$ and $T - 12g = 12a$ were seen. This

made it impossible for candidates to complete this part of the question.

In part (b)(i), some candidates were able to get back on track, as they had the correct equation for the 18 kg particle, but some did use the acceleration from part (a) instead of 3 ms⁻². In (b)(ii), there were many correct responses, but a few gave the answer 36 N, suggesting that there was some confusion between “reaction force” and “resultant force”. The responses to part (b)(iii) were quite mixed with some good solutions. It was not uncommon for candidates to simply assume that the friction was equal to 36 N. Other candidates tried to find the difference between a “tension” which was often incorrect and the 36 N.

There were a number of good answers to part (c), but some were too vague. For example just writing “particles” was not enough as it did not make the connection that it was the block that was being modelled as a particle.

Q17.

The force diagram asked for in part (a) was generally well drawn, but some candidates did lose marks for not including arrow heads or appropriate labels. The candidates who understood that they should resolve vertically to find the reaction force did well although some ended up with $R = 78.4 + 0.5T$ or equivalent. Others produced very confused responses.

Those who had a correct expression for the normal reaction often made some progress in part (c). There were quite a number of candidates who made promising starts but then made algebraic errors when solving for T . There were a range of errors that were made in setting up an equation of motion. These included only using the friction force, using T instead of $T\cos 30^\circ$ and omitting the ma term.

Q18.

Most candidates answered part (a) correctly but some just wrote down the printed result.

In part (a) some candidates started with $m \frac{dv}{dt} dv = -9.8v$ but found that this did not give the printed result, and just invented a term in g . This was not accepted. Some were penalised for ignoring the m term or for changing signs.

In part (b) candidates were expected to use $\int \frac{1}{v-5} dv = -1.96t$; some split $\int \frac{1}{v-5} dv$ into $\int \frac{1}{v} - \frac{1}{5} dv$. However there were many good answers to this part with both the integrations required being successfully completed. The difficulty which some candidates had was in the simplification needed to convert $\ln(v-5) = -1.96t + \ln 2$ into $v-5 = 2e^{-1.96t}$ which often became $v-5 = e^{-1.96t} + 2$

Q19.

Part (a) was done successfully by the vast majority of the candidates. The force diagrams in part (b) were usually good, but the most common errors were to omit either the normal reaction or friction forces. Also some candidates used lines rather than arrows to represent the force.

In part (c), the candidates were often able to produce the printed answer, but in some cases did not show enough working to justify the award of full marks. For questions like this, candidates should be advised to show an answer to more than three significant

figures before giving their final answer.

Those who started part (d) by resolving to find the friction force generally did well and completed this part of the question. A number of candidates did make significant errors by assuming, for example, that the friction force was $2g$ or in some cases $4g$.

Q20.

Some candidates produced very good clear solutions to this question. While there were many good force diagrams, some were confused and did not make use of arrows to indicate the forces. There were some issues with the direction of the tension and the 500 N force.

In part (b), the candidates who could start with a clear statement of the equation of motion, usually completed the question and gained full marks. For those who realised that an equation of motion was required, the most common errors were to omit a force or to take the wrong component. A few candidates omitted the ma term and treated the problem as if the forces were in equilibrium.

Q21.

This question was done well by many candidates. There were a few minor slips in part (a). Part (b) was a little more challenging. The most common errors were to omit one force, typically the weight leading to an answer of 160 N, or to make a sign error, for example $800g - T = 800 \times 0.2$ leading to a final answer of 7680 N.

Q22.

Parts (a), (b) and (c) were completed well by the majority of candidates and most of the errors were to be found in part (d). In part (a), some candidates made errors such as omitting arrowheads or using inappropriate labels, such as 4 kg instead of $4g$. In part (d), although there were many correct answers, the candidates produced a wide range of incorrect answers. Common errors included candidates' dividing either the 30 N (from the question) or 11.76 N (the friction) by the mass; other candidates calculated the resultant force and not the acceleration.

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Q23.

Parts (a), (b) and (c) of this question were done very well. In part (a), almost all candidates used an equation of motion for each particle and there were very few 'single equation' solutions. In part (c), some candidates stated one incorrect assumption, for example one about the peg rather than the string. In part (d), a few candidates took the acceleration to be equal to g rather than the value obtained in the question.

Q24.

Many candidates completed this question correctly. A common error occurring in part (a) was that some candidates in their diagram did not show that the two vertical upward reactions on the plank were different, or showed the two vertical gravitational forces acting at the same point. There were some interesting comments made in part (c), but most responses were correct.

Q25.

There was a range of responses to this question, with candidates generally doing better on parts (a), (b) and (c) than on part (d).

In part (a) the most common error was to either include or omit a negative sign in the early working, so that a negative time would have been produced if this had been followed through accurately.

In part (b) some candidates used an equation for the height rather than the velocity, but still equated it to zero. A few candidates ignored the initial height. Very few candidates realised that the ball would have the same speed when it returned to its initial position.

Q26.

This question was also done well by the majority of candidates. While most candidates were able to produce the printed answer for part (b), there were some issues with the force diagram and finding the tension in part (c). The two main problems with the force diagram were including extra forces, for example friction or simply poor diagrams without arrows or labels. In part (c) there were some confused responses and some candidates tried to resolve the normal reaction from part (b).

Q27.

Most candidates made good attempts at this question. In part (b) (ii) there were some solutions that did not contain enough working to justify the award of marks, for example a

$= \frac{4}{0.5} = 8$ with no explanation of where the 4 came from. Some also simply found a way to get 8, for example from something like $9.8 - 2 \times 0.9 = 8$.

For part (b) (iii) there were many correct solutions, but some candidates took the acceleration as 9.8 instead of 8. The final part proved to be more difficult for the candidates. While there were many candidates who produced good answers, there were also quite a lot who could not give a clear or relevant explanation.

Q28.

In part (a) many candidates lost marks for not including arrow heads or due to poor labelling of the forces. Several candidates did not show the two tensions as being equal on their diagrams. There were quite a few good attempts at part (b), with attempts at resolving vertically. Part (c) proved to be very challenging, although there were a number of good solutions. A number of candidates gained an M1 mark for the use of the friction law. The most common errors were to omit one force in their equation or to not include the ' ma ' term in their equation of motion.

Q29.

Many candidates completed this question correctly. A common error occurred in part (a) where some candidates did not show, on their diagram, that the two vertical reactions on the plank were different. Creative algebra by a number of candidates was seen in part (b), particularly in the insertion of ' g ' in the final answer.

Q30.

The candidates generally did well with parts (a) and (b), which were fairly standard, but had more difficulties with the later parts. The main reasons that candidates lost marks in part (a) were because they made errors with the signs in the equations or because they did not form an equation for each particle. Those with sign errors in part (a) often did not obtain the correct tension in part (b).

In part (c), many errors were due to the candidates using the wrong values in the wrong places. For example, in part (c)(i), some used an acceleration of 9.8 instead of 1.96 while in part (c)(ii), some used an acceleration of 1.96 instead of 9.8. Similar confusion concerned the use of the distance of 4 metres in part (c)(i) instead of in part (c)(ii), while the time of 2 seconds was sometimes used in part (c)(ii) instead of in part (c)(i).

Q31.

This question was also done well by many candidates. There were relatively few errors on the force diagrams. In part (b)(ii), some candidates made errors, the most common being to multiply the 30 by 0.5 or subtract the 30 from 49. Part (c) caused difficulties for some candidates. The most common errors here were to attempt to include the weight or to omit the friction force.

Q32.

The candidates generally did well with parts (a) and (b), which were fairly standard, but had more difficulties with the later parts. The main reasons that candidates lost marks in part (a) were because they made errors with the signs in the equations or because they did not form an equation for each particle. Those with sign errors in part (a) often did not obtain the correct tension in part (b).

In part (c), many errors were due to the candidates using the wrong values in the wrong places.

For example, in part (c)(i), some used an acceleration of 9.8 instead of 1.96, while in parts (c)(ii) and (c)(iii), some used an acceleration of 1.96 instead of 9.8. Similar confusion concerned the use of the distance of 4 metres in part (c)(i) instead of the later parts, while the time of 2 seconds was sometimes used in the later parts instead of in part (c)(i).

In part (c)(iii), the most common approaches involved the use of constant acceleration equations. While some candidates used the correct numerical values in their equations, many made errors with the signs. Another common approach was to find two times and add them together. With this method also, sign errors were often made by the candidates.

Q33.

There were many good solutions to part (a). Only a very small number of candidates ignored the instruction to form an equation of motion for each particle. A few candidates assumed that the heavier particle would accelerate upwards when released. Some of these candidates worked through to produce a negative acceleration and stated that magnitude would be the required value. However, some of the candidates who took this approach “lost” a negative sign in their working and ended up with a positive acceleration.

In part (b) there was one common error, which was seen on a fairly large number of scripts. This was to obtain the distance moved by each particle, but not to double this to give the distance between the particles. A very small number of candidates used an

acceleration of 9.8 instead of 0.98.

Q34.

The printed answer in part (a) seemed to help the candidates as there were very few incorrect answers to this part of the question. In part (b), while there were many good

responses, some candidates used $\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{0.9}{0.6} = 1.5$ and then applied the equation $v = u + at$ to obtain an acceleration of 2.5. There were some very good answers to part (c), but some candidates gave answers such as “The accelerations are different.” but did not explain why. In the better explanations, candidates made clear references to friction forces or air resistance.

Q35.

Parts (a) and (b), about the assumptions, were generally answered well. The most common incorrect answer was to say that the peg was fixed in part (a).

There were many good solutions to part (c). Only a very small number of candidates ignored the instruction to form an equation of motion for each particle. A few candidates assumed that the heavier particle would accelerate upwards when released. Some of these candidates worked through to produce a negative acceleration and stated that magnitude would be the required value. However, some of the candidates who took this approach “lost” a negative sign in their working and ended up with a positive acceleration.

In part (d) there was one common error, which was seen on a fairly large number of scripts. This was to obtain the distance moved by each particle, but not to double this to give the distance between the particles. A very small number of candidates used an acceleration of 9.8 instead of 0.98.